

Sustainable paths to food security from the perspective of food loss and waste management

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Abstract—Food security crisis has been accelerated by the Russia-Ukraine war globally. One-third of the world's produced food is lost and wasted annually. Food loss and waste (FLW) is about food security, safety, quality, and sustainability. The motivation for undertaking this comprehensive review is to provide useful information and directions for government, stakeholders, public organizations, and individuals to take action and develop practices on minimizing food loss and waste for the benefit of society, the environment, and the economy. In this review, we used secondary research review and content analysis as a research methodology to analyze and simplify the sustainable path of food security from the perspective of food loss and waste management. In the end, we found it is prospective to pursue sustainable food security by way of reducing food loss and waste.

Keywords—food loss, food waste, waste management, consumer behavior

I. INTRODUCTION

Food loss and waste is not only a concern of food security but also food safety, food quality, and sustainability. According to the data of FAO in 2020, more than 811 million population were in hunger, and more than 3.1 million population were not able to access a healthy diet. Due to the impact of the COVID-19 pandemic, more than 132 million people are suffering from food and nutrition insecurity [1], [2]. Besides, the war outbreak in Ukraine (which started in Feb. 2022 and still going on at the end of Dec. 2022) just simply accelerates the global food security alarm [3]–[5] because the two conflicting countries, Russia and Ukraine, are very important world foodstuff suppliers and fertilizers suppliers. And Russia is also one of the most important oil and gas suppliers [6]. Especially the war between Russia and Ukraine disturbed the regular and important function of the “world bread basket” around the black sea area [3], [7]. Besides these unpredicted human catastrophes, such as the COVID-19 pandemic and war in Ukraine, there are some consistent threats and risks in agriculture and food. For example, chronic climate change or extreme weather (threaten agriculture through their influence on ecology, the environment, the geographical situation of crop and crop production, the resources and supply chain of agriculture, and the market price) [8]–[10], natural resources scarcity (arable lands and water) [9], [11], agriculture facilities issues (aging farmers and fewer farmers because of urbanization) [11]–[13], food market fluctuation [14]. Pursuing a sustainable path of food security is not only a feasible solution but also an emergent path for our present and future generations.

Therefore, the aim of this research is to provide a comprehensive review to simplify the sustainable path of

food security based on the nexus between food security and food loss and waste.

II. METHODOLOGY

In 1993, Nyambo B.T. was the first author to establish a link between food loss and food security issues [15]. So far, the state of art research approaches on the topic are Systematic and bibliometric approaches. In this review, we used literature review [16] and content analysis [17] as the research method to study the sustainable paths to food security and the solutions for reducing food loss and waste along the value chain of agriculture. This research methodology helped to provide a comprehensive understanding of the current status of food loss and waste and point the future direction on the related topic.

III. RESULTS

A. Basic definitions

1) Food loss and food waste

Unfortunately, the two concepts “food loss” and “food waste” are usually used interchangeably, which is not true [18], [19]. In a short sentence to describe food loss and waste: decrease in the quantity and quality of food across the food value chain [20]. Food loss and waste happens all the along the food value chain [21]. Food loss is mainly happening upstream of the food value chain before the last stage, “consumption”, such as agricultural production (cultivation, breeding, harvest/slaughter/catch), industrial processing, and distribution (**Figure 1**) [22].

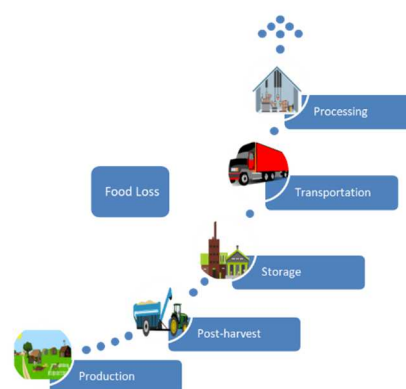


Figure 1. Source of food loss across the food value chain

Source: own construction

Similarly to the definition of food loss, food waste refers to the decrease in quantity and quality of food at the consumption stage across the entire food value chain, such as food service, retail, household [23], and other formats of consumption (**Figure 2**). However, only the agricultural

and breeding products (plants and animals) and their fragments and the final edible food intended to be for human consumption are classified into food loss and food waste. The difference between quantitative and qualitative food loss and food waste is explained in

Table 1.



Figure 2. Source of food loss across the food value chain

Source: own construction

Table 1. The difference between quantitative food loss and food waste and qualitative food loss and food waste

	Feature	Quantitative Stage	Participants	Feature	Quantitative Stage	Participants
Food Loss	Mass of food is removed	Upstream	Farmer, Processor, Distributor	Food value is decreased	Upstream	Farmer, Processor, Distributor
Food Waste	Mass of food is decreased	Downstream	Retailer, Consumer, Food service provider	Food value is decreased	Downstream	Retailer, Consumer, Food service provider

Source: Own construction

2) *Surplus food*

Those food from farm to fork are still edible and safe for human consumption, but behind the aesthetic specifications or fail to sell before the date labeled “Best Before” are becoming surplus food [24]. Surplus food is also generated from overproduction because there is no efficient “demanding” communication between food producers and the market and the damaged packaging [25]. Surplus food will end up as food waste if it is not redirected for human consumption, which also causes energy loss, water, and other natural resources loss, and generate greenhouse gas. However, surplus food is still food, just like a baby but with bathwater [26]. We should not throw surplus food, just like we cannot throw a baby because it has bathwater. It is important to redirect surplus food for human consumption, which helps to strengthen our community.

3) *Food supply chain and food value chain*

The food supply chain means the process from farmers (suppliers) to market (distributor) [27]. The food supply chain is more about logistics. According to Michael E. Porter’s value chain theory [28], value chain refers to the full chain of a business’s activities in the creation of a product or service--- from the initial reception of materials all the way through its delivery to market, and everything in between. The food value chain has five primary activities (inbound operations, operations, outbound logistics,

marketing and sales, and service) and four secondary activities (company infrastructure, human resource management, technological development, procurement, and purchasing).

4) *Questions about the logistics of food*

Food logistics concentrates on the food supply chain process, such as the very important points [29]:

- Transportation of goods
- Vehicle selection to meet specific requirements to transport goods
- Materials and tools selection to handle goods
- Proper storage
- Inventory management

The food supply chain or logistics of food plays a significant role in ensuring the smooth flow of distributing good quality and safe food to humans [30]. An effective logistic model can not only strengthen food distribution but also reduce food loss and waste and lessen the impact of food loss and waste [31].

B. *Sustainable path tackling FLW across the agriculture value chain*

1) *At the agricultural primary production and storage stage*

The main cause of food loss at the primary production stage is: moisture, grain damage, and pest outbreaks, and the less frequent cause are: overgrowth, mouldiness, change in grain odor, or occurrence of fire in the warehouse. The feasible ways of eliminating grain loss at the primary production stage can be composted, used for animal feeding, used as raw materials for biofuel, used for biogas plants, or buried in landfill [18]. On the other hand, the proper packaging method [32] (use air-tight or hermetic bags instead of porous woven polypropylene bags) can help to reduce the possibility of grain diseases and pesticide damage during the storage stage [33]. The application of dynamic controlled atmosphere technologies in apple storage and controlled atmosphere conditions for pears can maximum keep the physical appearance. Cold storage is also important to prevent the maturity of perishable fruits, meat, and vegetable. The freeze-drying method is proven as a better method than hot-air drying for food preservation. Future grain storage [34] and other agri-food product storage should focus on more high-tech equipment and intelligent methods. Besides, precision technology can also make primary production more effective [35], such as bridging the production information asymmetry and reducing its vulnerabilities [36]. A short supply chain also contributes to the local circular economy and reduces food loss and waste [37].

2) *Post-harvest processing stage*

Sometimes food is lost because of failure to meet the systematic standards. Reprocessing unwanted food from the food supply chain can avoid food loss [19]. For example, those discarded eggs that are a little bit dirty, a little bit broken, or out of classification can be reprocessed into egg products. The fruits, such as apples and mango, have a little bit of mechanical injury or are out of shape and can be reprocessed into jams, preserves, canned slices,

dried slices, and so on [38]. The discarded dairy and meat from technical issues, unfavorable management, improper packing, and so on can be reprocessed as well. Reprocessing the discarded food method can also be practiced into fruits and ready-to-eat meals or instant foods. Using built-in active components active packaging can reduce food waste by extending food product shelf life, which is also environmentally friendly by replacing preservatives [39]. Recycling discarded food (human consumption) in line with animal feed requirements is also a regular measurement for reducing food loss and waste. Due to the immature food processing market, the interventions from the government can help to ease the supply chain challenges, such as initiatives, regulations, and taxation policy, more acceptance attitude to the novel and innovative processing products and processing methods, and comprehensive safety and quality management system [38]. Derivatives of the primary food products, such as mango pulp, can also be processed as the raw material to enrich and flavor secondary products (yogurt, bakery, soft drink, ice cream, and so on). Generally, the byproducts (kernel and seeds of fruits account for 20-60% [40] and peel accounts for up to 30% [41] of fruit weight) are a source of waste. Byproducts of primary food products still have rich bioactive compounds, and they can be manufactured again as value-added and edible food products, such as food fortification [42] and feeds. For example, there are plenty of carotenoids, polyphenols, and dietary fibers in mango's byproducts, peel, and kernel [38]. The kernel can be processed into oil (15% edible oil from mango kernel, 18-20% oil from soybeans and cotton seeds) [40] and flour [43]. The peel can be used as functional food [44], replacement in cream, yogurt [45], and cheese products, and as flavoring ingredients in food and cosmetic products [46]. Strengthening the awareness of employees about the consequences and the importance of human error in food processing plays an important role in food loss during the food processing stage [47]. The food lost from the processing stage can also be used to feed animals, compost, and for industrial use, such as bakery and confectionery [48].

3) Transform food waste into high-value-added products

Instead of landfills, using transformation technologies to reuse food surplus or food waste and create value-added products is an economic efficiency and sustainable environmental solution. For example, bread waste is an increasingly global issue [49], especially due to the price crisis in post COVID era and the ongoing Russia-Ukraine war [50]. Bread waste is happening at all stages of the food value chain from industrial processing. But bread waste and other food byproducts [42] could be transformed as high value-added products in chemistry, materials, nutrient-enriched products, and bioenergy products such as:

- Fuel source: ethanol
- Energy source component: hydrogen
- Organic acid which is applied in food, beverage, cosmetics, pharmaceutical, chemistry, and so on: lactic acid

- Precursor molecule for chemical compounds synthesis used in food, pharmaceutical, polymers, and so on: succinic acid
- Resources for enzyme production as a catalyst in chemical reactions
- Renewable biomass: can be developed in hydroxymethylfurfural synthesis
- Protein and pigment
- Fermentable sugar: glucose-fructose syrup
- Aroma compounds production: waste bread can be fermented to get aroma compounds
- Grow media for microorganisms

Besides the above-mentioned high-value-added products from food waste and food byproducts, in light of the waste-energy-water nexus (because all food production needs energy and water), food waste can also be utilized for biofertilizers and nanomaterials [51]. Another example is corn byproducts reprocessing and utilization, such as corn gluten meal, corn husk, steep corn liquor, corn germ, and fuel ethanol [52]. In the future, the proposed technologies should be considered deeply to accelerate the circular economy by transforming food waste into a bioresource and other high-value-added products.

4) Extend products' shelf life

Date labeling is one of the significant reasons for food waste at the consumption stage. Especially for perishable food products, such as fruits, vegetables, and meat, the short shelf-life is challenging food security. There are plenty of solutions to extend products' shelf-life, such as developing innovative and safe food packages. For example, at Inner Mongolia Agricultural University, a water-absorbent anti-bacterial mat was developed for fresh mutton to extend its shelf-life up to eight days at 4°C storage periods [53]. Anti-bacterial pads containing cinnamaldehyde can also extend cold fresh pork quality and shelf-life [54]. And it has been found that PLA - lemon essential oil nanofibrous membranes as an anti-bacterial material can extend Mongolian cheese by two days [55].

5) Innovative selling methods

Selling unsold but good quality, intact and fresh food at a specific time period or end of the opening day helps restaurants, bakeries, supermarkets, hotels, and other stores save money from surplus food and helps consumers to save money as well. For example, MunCharity is a joint charity project between Munch and the Hungarian Food Bank. People can order food at a 40-70% discount via munch.hu platform from a browser or APP [56], [57]. Or selling surplus food as a food blind box at a lower price is also a novel selling strategy to avoid food waste [58] and selling via e-commerce is also a sustainable solution [59]. But all these innovative selling methods need to be improved their reputation, attract consumers' purchase and decrease risks.

C. Sustainable path tackling FLW from the perspective of public and individual

1) Strengthen policy and education investment

Strengthening the policy, initiatives or regulations is important not only at a national level but regional and

international cooperation. As an important sector of the EU economy, the agri-food chain guarantees the service and income source for Europeans. In order to have sustainable food production and food consumption, at the same time reducing food waste, the European Commission adopted Farm to Fork Strategy as its key action of the European Green Deal in 2020 [60]. This strategy has the mission to shift the European food system into a sustainable mode by 2050. Food security is the priority of this strategy, and the following as well:

- ensure sufficient, affordable, and nutritious food within planetary limits
- halve the use of pesticides and fertilizers and sales of antimicrobials
- increase the amount of land devoted to organic farming
- promote more sustainable food consumption and healthy diets
- reduce food loss and waste
- combat food fraud in the supply chain
- improve animal welfare

Reducing food waste is one of the three initiatives and legislative proposals (the other two are organic farming and front-of-pack nutrition labeling, and sustainable food labeling). Similar to Farm to Fork Strategy, 2011 The Roadmap to a Resource Efficient Europe [61] also pointed out that “reduced food waste can contribute to improving resource efficiency and food security at a global level” [62], [63].

Besides addressing the national or regional strategies for the reduction of food loss and waste, necessary law enforcement is also needed in some cases. In 2016, France carried out a law banning retailers from discarding unsold food which is fit for consumption [19]. “Food Donation” is a good echo to the law, which means the retailers (the main donors), industries, and farmers donate unsold food that is still fit for consumption to the food supply chain. The most common donated food categories are those that need less processing before consumption, such as fruits, vegetables, and eggs.

In addition to policies and regulations on controlling food loss and waste, education or training on reducing food loss and waste and sufficient operation are also important as the entire food value chain is humanely involved [64]. Control and educate farmers, workers, employees, retailers [65], and consumers aware of the consequences of human error in operations [47] and the importance of reducing food loss and waste, and the practical technologies and skills, such as packaging [33], are a very important step to reduce food loss and waste. Nutritional education and healthy lifestyle education for the general population are also a solution for food loss and waste reduction solution [65].

2) *Increasing consumers' consciousness on reducing food waste*

The first step to reducing or avoiding risks is to perceive risks. Awareness campaigns were investigated as having a positive influence on consumers' limit of food waste among 384 respondents in 2020 in Romania [66].

Food loss and waste at the consumption stage account for huge weight across the entire food value chain. According to Roger's Diffusion of Innovation Theory (DOI), the crucial step to make people execute the implementation of food loss and waste reduction is to make people perceive it [67], [68]. Increasing consumers' or human's awareness of food loss and waste [69] should focus on the ages population since childhood or primary school [70]. Digital education [71] can be a way to cultivate people's healthy and wise eating behavior and habits without distance, time, and accessibility limitations. The built of International Day of Awareness of Food Loss and Waste [72] by FAO and UNEP aimed to raise people's awareness of working for a sustainable food system by 2030. In short, education is a very crucial solution to raising people's awareness of reducing food loss and waste. And Persuading consumers to buy local food via educational campaigns can also help to reduce food loss and waste [73]. Especially after COVID-19, people prefer to buy domestic food in some countries, such as Hungary [74], [75]. At the same time, propaganda, such as international day or activities, is also a practical way to attract consumers' attention.

Besides, the role of some organizations coordinating multiple players in the food value chain can also be useful in reducing food waste. An instance of success in Hungary involves local food-buying clubs (referred to as BCs) [76]. These organizations are driven by consumers, and their goal is to connect local food producers with consumers. The main goals are to show support for local producers by paying fair prices instead of low ones, boost the local economy, and acquire healthy, safe, and dependable food with minimal environmental impact.

IV. CONCLUSION

The sustainable path to food security is not only guaranteed by the source of producing food (such as the modern and sophisticated smart agricultural technologies as a lot of valuable literature mentioned), but also by the awareness and knowledge of utilizing the food resources more wisely and efficiently across the entire food value chain (as it is explored and highlighted in this research, reducing food loss and waste is contributing to a more sustainable food system and a stronger community). In order to realize a sustainable food security and food system, all the public and private sectors should have a joint effort on the way, strengthen consumers' awareness of reducing food waste, and adopt the “degrowth” theory in agricultural production and food consumption [77], [78], especially the support from policies and investment in education. We also highlight the future research direction for other passionate researchers focusing on evaluating food loss and waste reduction measurements or technologies [79], which will further help all the players pursue sustainable food security.

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